

Phase 3 Diagnostic Iteration: Topological Auto-Covariance Analysis

Antigravity AI Pipeline

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1 Diagnostic Objective

Pursuant to the Phase Gating protocol, the ECCM Mixed Model failed Phase 2 Calibration against the WSLS baseline. We halted scaling to project the scalar residuals back onto the Granule Cell (N_{GC}) layer and map the temporal prediction error auto-covariance.

2 Mathematical Diagnostics

- **Temporal Error Auto-Covariance:** The ACF of ϵ_t reveals **1** significant correlation spikes beyond lag 0. This mathematically proves a *Temporal Delay Shortage*.
- **Granular Spatial Projection:** The condition number of the Granule Cell covariance matrix is **699719.73**. This indicates extreme multicollinearity and *Spatial Collision* within the Granular Layer.

3 Root Cause Analysis

The Cerebellar Mossy Fibers ($N_{MF} = 20$) currently operate as a rigid, discrete shift-register, abruptly cutting off history after 10 trials. The environment contains macroscopic probabilistic reversals (40-80 trials). The sudden loss of history at trial 11 causes the Granular projection to mathematically collide, resulting in temporal delay shortages. The WSLS baseline defeated the ECCM because WSLS uses an exponential leaky integrator, which possesses effectively infinite memory depth.

4 Topological Modification Proposal

To resolve this structural failure without arbitrarily altering the 20:132:170 Information Bottleneck, we must upgrade the Mossy Fiber Topology.

Current Architecture (Discrete Shift Register):

$$MF_i^{(t)} = Input^{(t-i)}$$

Proposed Architecture (Continuous Leaky Integrator Basis): We transform the Mossy Fibers into a spectrum of Leaky Integrators with logarithmically spaced time constants τ_i :

$$MF_i^{(t)} = \left(1 - \frac{1}{\tau_i}\right) MF_i^{(t-1)} + \left(\frac{1}{\tau_i}\right) Input^{(t)}$$

Where $\tau_i \in \{2, 4, 8, 16, 32, 64, 128, \dots\}$.

This topological modification mathematically resolves the temporal delay shortage by embedding infinite, decaying historical context directly into the Mossy Fibers, eliminating Spatial Collision.